

## Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

## Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$ ,  $x_d[k]$  is the desired motion trajectory,  $\delta x_D[k]$  is the motion error caused by the disturbance, and  $\Delta x_d[k] = x_d[k+1] - x_d[k]$  is the desired motion step size.

## Measurement

( $d$ -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

**Control objective:**  $\delta x[k+1] = \lambda_c \cdot \delta x[k]$ ;  $0 \leq \lambda_c < 1$

*Ideal control effort (no measurement delay):*

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta \mathbf{x}[\mathbf{k}] - \delta x_D[k]\}$$

*Motion error:*

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

*$d$ -step measurement delay:*

$$\delta \mathbf{x}[\mathbf{k}] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

*Ideal control effort (with  $d$ -step measurement delay):*

$$\begin{aligned} \bar{\bar{f}}_{dx}[k] = a_x^{-1}[k] \Big\{ & \underbrace{\left[ \Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[ \delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \\ & - \underbrace{\left[ \delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \Big\} \end{aligned}$$

*Motion error:*

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i]f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

## Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[ \delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

### Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_T^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[ \delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

$$a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[ (\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

### Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

$$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k] \Delta h_d[k], \quad \Delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k]$$

$$\Delta h_d[k] = h_d[k+1] - h_d[k] \text{ (desired-trajectory height increment).}$$

$$\text{Parameter estimator:} \quad \hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k]$$

$$\text{Parameter estimation error:} \quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$\text{Estimation error dynamics:} \quad e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k]$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \sum_{j=1}^i \Delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{f}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \Delta a_{ram}[k-i] \right\}}_{\delta x_{\delta a_f}^d[k]} \\ &\quad + \delta x_D^d[k] + (1 - \lambda_c) n_x[k] \end{aligned}$$

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x_D^d[k] - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta a_f}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]}$$

### State equation

( $\delta x_D^d \equiv 0$ : disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k]$$

### Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^d \Delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \Delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \Delta a_x[k-i] \end{bmatrix} \text{ (known)}, \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

### Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k], \quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = e_{\delta x_1}[k] + n_x[k], \quad e_{y_2}[k] = y_2[k] - \hat{y}_2[k] = e_{a_x}[k] + r_2[k]$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

### Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k]$$

$$e_{\delta x_3}[k+1] = \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]}$$

$$e_{a_x}[k+1] = e_{a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \underbrace{\Delta a_{ram}[k]}_{q_4[k]}$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k - i]$$

$$\mathbf{e}[k + 1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$

**Gain fluctuation variance**  $\text{Var}(\Delta a_{ram})$

**Assumption** ( $x = x_{det} + x_{ram}$ , perfect tracking  $x_{det} = x_d$ ):  $\delta x = x_d - x = -x_{ram}$  (zero-mean),  $x_{ram} = -\delta x$ .

$$a'_x := \frac{da_x}{dh} = -\frac{a_x K_h}{R}, \quad K_h = \frac{1}{c} \frac{dc}{d\bar{h}} \text{ (wall sensitivity)}, \quad a_{x_{ram}} = a'_x x_{ram}, \quad x_{ram} = -\delta x$$

**Closed-loop tracking variance** (thermal-driven,  $\hat{a}_x = a_x$ ):

$$\text{Var}(x_{ram}) = \text{Var}(\delta x) = C_{\delta x} \sigma_{\delta h}^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}, \quad \sigma_{\delta h}^2 := 4k_B T a_x$$

**Increment of a stationary signal** ( $\rho_1 = \text{Cov}(x_{ram}[k+1], x_{ram}[k]) / \text{Var}(x_{ram})$ , lag-1 autocorr):

$$\text{Var}(\Delta x_{ram}) = 2 \text{Var}(x_{ram}) - 2 \text{Cov}(x_{ram}[k+1], x_{ram}[k]) = 2(1 - \rho_1) \text{Var}(x_{ram})$$

$$\rho_1 = \lambda_c + \frac{2(1 - \lambda_c)}{C_{\delta x}} \Rightarrow 1 - \rho_1 = \frac{1}{(1 + \lambda_c) C_{\delta x}}$$

**Gain increment**  $\Delta a_{ram} = a'_x \Delta x_{ram}$  ( $C_{\delta x}$  and  $(1 - \rho_1)$  cancel):

$$\text{Var}(\Delta a_{ram}) = a_x'^2 \text{Var}(\Delta x_{ram}) = a_x'^2 \cdot \underbrace{2(1 - \rho_1) C_{\delta x}}_{= 2/(1 + \lambda_c)} \sigma_{\delta h}^2$$

$$\text{Var}(\Delta a_{ram}) = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_{\delta h}^2$$

**Impl.:**  $K_h$ ,  $\sigma_{\delta h}^2 = 4k_B T a_\perp$  evaluated at  $\bar{h}_{\text{meas}}$  (measured,  $p_m$ ); gain  $a_x \rightarrow \hat{a}_x$  (estimate).

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**Correction —  $a_{x_{ram}}$  is first-order autoregressive (AR(1)), not a random walk**


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**True-signal dynamics** (no estimate;  $x_{ram} = -\delta x$ ,  $\delta x[k+1] = \lambda_c \delta x[k] - \varepsilon[k]$ ):

$$a_{x_{ram}}[k+1] = -a'_x \delta x[k+1] = -a'_x (\lambda_c \delta x[k] - \varepsilon[k]) = \lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]$$

$$\Delta a_{ram}[k] = a_{x_{ram}}[k+1] - a_{x_{ram}}[k] = \underbrace{(\lambda_c - 1) a_{x_{ram}}[k]}_{\text{reversion}} + \underbrace{a'_x \varepsilon[k]}_{\text{white}}$$

**True gain state** ( $a_x = a_{x_{det}} + a_{x_{ram}}$ ,  $a_{x_{det}}[k+1] = a_{x_{det}}[k] + \Delta a_x[k]$ ):

$$\begin{aligned} a_x[k+1] &= (a_{x_{det}}[k] + \Delta a_x[k]) + (\lambda_c a_{x_{ram}}[k] + a'_x \varepsilon[k]) \\ &= (a_{x_{det}}[k] + \Delta a_x[k]) + \lambda_c (a_x[k] - a_{x_{det}}[k]) + a'_x \varepsilon[k] \\ &= \underbrace{\lambda_c a_x[k]}_{\mathbf{F}_e(4,4)=\lambda_c} + \underbrace{(1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k]}_{\text{known input}} + \underbrace{a'_x \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$\mathbf{F}_e(4,4) = \lambda_c$

**Estimator** (predict reverts to  $a_{x_{det}}$ ):

$$\hat{a}_x[k+1] = \lambda_c \hat{a}_x[k] + (1 - \lambda_c) a_{x_{det}}[k] + \Delta a_x[k] + \ell_{4\cdot}[k] e_y[k]$$

$$\begin{aligned} e_{a_x}[k+1] &= a_x[k+1] - \hat{a}_x[k+1] \\ &= \lambda_c (a_x[k] - \hat{a}_x[k]) + a'_x \varepsilon[k] - \ell_{4\cdot}[k] e_y[k] \\ &= \lambda_c e_{a_x}[k] + a'_x \varepsilon[k] - \ell_{4\cdot}[k] e_y[k] \end{aligned}$$

**Process noise** (white driver  $w_4 = a'_x \varepsilon$ ;  $\text{Var}(\varepsilon) = (1 - \lambda_c^2) \text{Var}(\delta x)$  since  $\delta x$  is AR(1)):

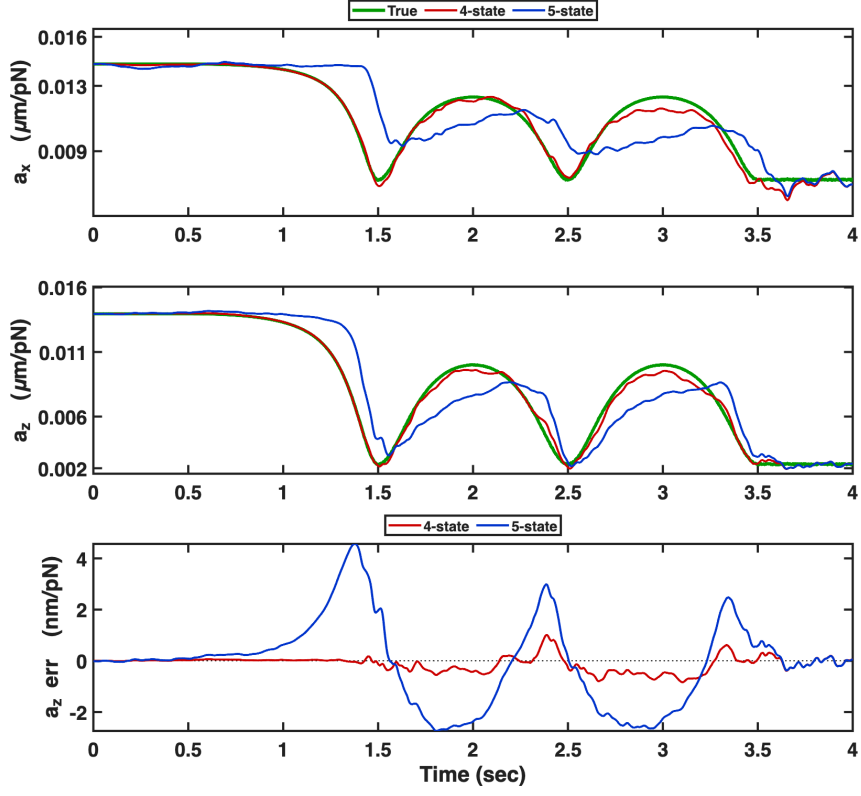
$$q_4 = \text{Var}(w_4) = a_x'^2 \text{Var}(\varepsilon) = a_x'^2 (1 - \lambda_c^2) \text{Var}(\delta x) = (1 - \lambda_c^2) C_{\delta x} a_x'^2 \sigma_{\delta h}^2$$

$q_4 = a_x'^2 \text{Var}(\varepsilon) = (3 - 2\lambda_c^2) a_x'^2 \sigma_{\delta h}^2$

**Impl.:** predict and  $q_4$  use  $\hat{a}_x$  with  $a'_x$  at  $\bar{h}_{\text{meas}}(p_m)$ ; reversion target  $a_{x_{det}}$  from  $p_d$ .

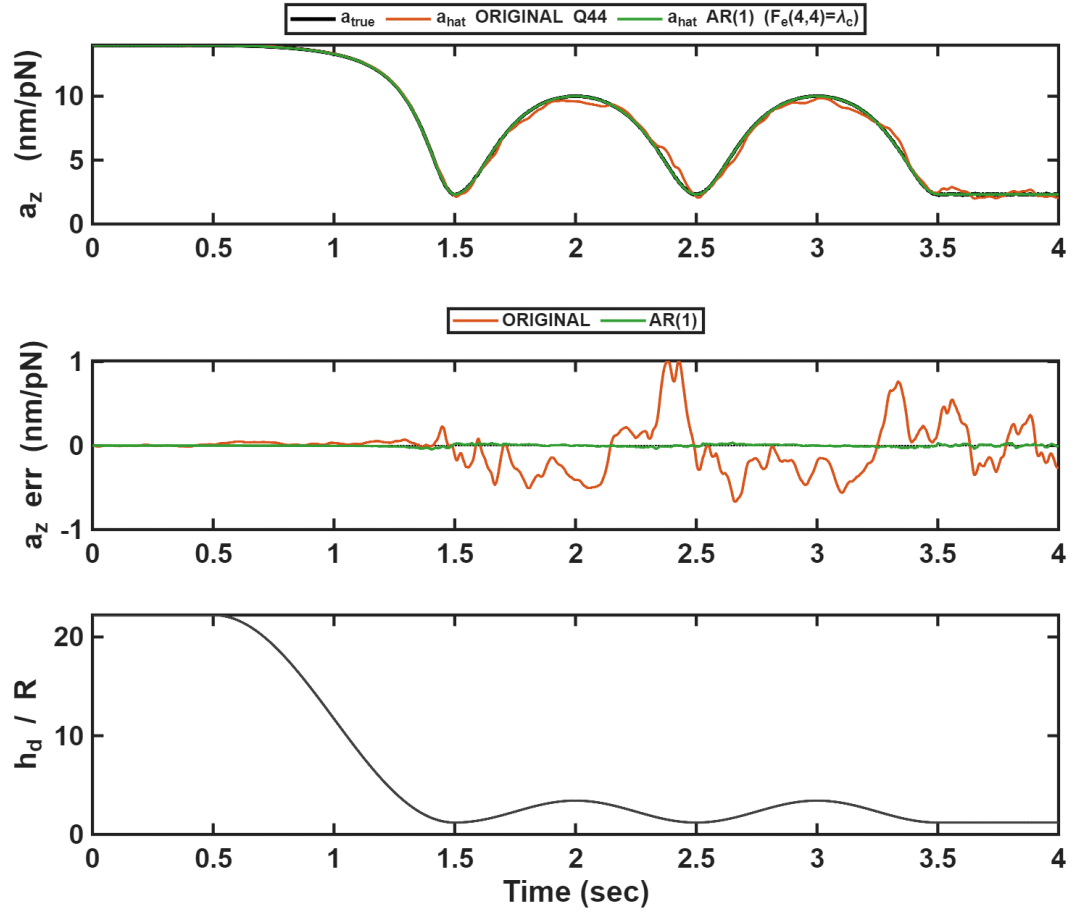
### Verification (1 Hz, near wall)

Per-axis EKF, seed ensemble; near-wall 1 Hz oscillation ( $h_{init} = 50 \mu\text{m}$ , trough  $\bar{h} = 1.2$ ).



$a_x, a_z$ : true gain and the 4-state / 5-state estimates; bottom tile, the  $a_z$  estimation error  $\hat{a}_x - a_x$ .





$a_z$ : original random-walk  $Q_{44}$  ( $F_e(4,4) = 1$ ) vs  $AR(1)$  reverting gain ( $F_e(4,4) = \lambda_c$ ); middle tile, estimation error; bottom,  $\bar{h}$ . Near-wall  $|\hat{a}_z - a_z|$ : 9.4%  $\rightarrow$  0.4%.